

Pipeline Discovery: Empirical Structural Classes

1. Emergence of Type-Signature Classes (σ)

The pipeline discovered 4 distinct type-signature classes (σ) emerging from the 8 analyzed `program.md` files:

Class	σ (Sections Present)	Members
σ -01	{setup, scope, output, logging, loop, meta}	S1-RF-1, S1-RF-2, S1-RF-minimal, S2-GBM-1
σ -02	σ -01 + {eval, strategy}	S1-RF-5, S2-GBM-5
σ -03	σ -01 + {strategy}	S1-RF-4
σ -04	σ -02 + {budget}	S3-RF-HP

This is a substantive finding: **S1-RF-1** and **S2-GBM-1** land in the exact same σ -class. Although they represent different algorithms (RF vs GBM), their narrow-terse variants are structurally identical at the coarse level. If the autoresearch behaves differently on them, that difference cannot be explained by the type-signature; it must stem from either feature magnitudes (the $\sim_\phi$ level) or semantic content discarded by the AST (the $\sim_T$ level).

2. Feature Singletons and the ε -Threshold

At a tolerance of $\varepsilon = 0$, all 8 files are ϕ -singletons. This is expected, as every file possesses at least one unique feature count.

The key scientific progression for this empirical study is to **increase ε gradually** until pairs merge into equivalence classes, and then verify whether these merged pairs also produce similar behavioral outcomes in Phase 2.

3. Tree-Isomorphism and ZSS Distance

The Zhang-Shasha (ZSS) distance matrix reveals the cleanest structural relationships across the ASTs:

- **Distance 0 (Isomorphic):** S1-RF-1, S1-RF-2, and S2-GBM-1 are tree-isomorphic modulo leaf text.
- **Distance 1 (Near-Isomorphic):** S1-RF-5 and S2-GBM-5 differ only by one feature leaf.
- **Degenerate Variant:** S1-RF-minimal operates as a near-zero-distance variant of the σ -01 class, serving as an ideal control to test the verbosity floor.

